

General GK, English & NIELIT Patterns

This note is designed for rapid revision of general awareness, English spelling rules, and specialized NIELIT exam patterns (such as Scientist B and Scientific Assistant) before stepping into the exam hall.

[STPI] Part 0: NIELIT Core Exam Patterns & Solved PYQs

Unlike TCS or IBPS, NIELIT-conducted exams are heavily influenced by the GATE computer science syllabus, focusing on theoretical concepts, mathematical proofs, and system-level behaviors.

1. Theory of Computation (TOC) & Automata

Phase-Structured Grammar: Unrestricted Grammar (Type-0) is also known as Phase-Structured Grammar. It is the most general grammar.

Turing Machine Equivalences:

The power of a multi-tape Turing machine is equal to a single-tape Turing machine (it only increases speed, not power/capability).

Every Non-deterministic Turing Machine (NDTM) has an equivalent Deterministic Turing Machine (DTM) of equal power. Determinism = Non-determinism holds true for Turing Machines and Finite Automata, but fails for Pushdown Automata (NPDA & DPDA).

Finite State Automata Limits: Palindromes and arbitrary memory strings ($a^n b^n$) cannot be recognized by a Finite State Automaton because it lacks arbitrary memory storage.

2. Compiler Design

Parser Power & State Relations:

Power order: $\text{CLR} \geq \text{LALR} \geq \text{SLR} \geq \text{LR}(0)$.

State size relationship: If SLR has n_1 states, LALR has n_2 states, and CLR has n_3 states:

$$n_1 (\text{SLR}) = n_2 (\text{LALR}) \leq n_3 (\text{CLR})$$

Hence, SLR and LALR have exactly the same number of states because LALR merges common-core states of CLR.

Compilation Phases:

Lexical Analysis (output = tokens) $\rightarrow$ Syntax Analysis / Parsing (output = parse tree) $\rightarrow$ Semantic Analysis (type checking) $\rightarrow$ Intermediate Code Generation (AST/three-address code) $\rightarrow$ Code Optimization $\rightarrow$ Code Generation.

Trap: Type checking is done after parsing (during Semantic Analysis), not before!

Parser Conflicts:

Shift-Reduce Conflict: Arises in LR parsers when the parser cannot decide whether to shift the next input symbol or reduce the current stack elements using a grammar rule. Often caused by grammar ambiguity.

Shift-Reduce Parsing is also called the Reverse of Right Most Derivation (RMD in Reverse).

3. Databases & Transactions

Strict Two-Phase Locking (Strict 2PL): Generates schedules that are both conflict-serializable and recoverable (and cascade-less) because write locks are not released until transaction commit/abort.

Thomas' Write Rule: Applied to Timestamp-Ordering protocols. It ignores out-of-date write operations (blind writes) and can generate view-serializable schedules that are not conflict-serializable.

4. Computer Networks

Class B Subnetting Calculations:

In Class B: Default Net ID = 14 bits, Host ID = 16 bits.

Default number of networks = 2^{14} . Default hosts per network = $2^{16} - 2$.

If a problem states: "Class B network has 2^m networks and $2^n - 2$ hosts," the relationship is:

$$m = 14, n = 16 \implies 8m = 7n \implies (8 \times 14 = 7 \times 16 = 112)$$

To divide Class B for 64 departments: $64 = 2^6$ implies steal 6 bits from Host ID. The subnet mask becomes 255.255.252.0 (since $255.255.(11111100)_2 = 255.255.252.0$).

TCP Header & Congestion:

TCP Checksum: Calculated over the TCP Header, TCP Data, and a 12-byte Pseudo-Header (which contains Source IP, Destination IP, Protocol, and TCP Segment Length). Calculated using 16-bit one's complement sum.

Congestion Avoidance: The congestion window increases additively ($+1 \text{ MSS}$ per RTT) during avoidance and decreases multiplicatively (halved) upon detecting packet loss via triple-duplicate ACKs (or drops to 1 MSS on a Timeout).

Part 1: General Knowledge & Current Milestones (2026 Updates)

1. India's Historic Nuclear Energy Milestone (April 2026)

What: Achieving first criticality (self-sustaining controlled nuclear fission chain reaction).

When: 6th April 2026 at 08:25 PM.

Where: Prototype Fast Breeder Reactor (PFBR) at the Kalpakkam Nuclear Complex, Tamil Nadu.

Significance:

Officially transitions India into Stage-2 of its three-stage nuclear power program first envisioned by Dr. Homi Jehangir Bhabha.

PFBR is a 500 MWe (MegaWatt electrical) sodium-cooled fast breeder reactor.

Built entirely using indigenous technology by BHAVINI (Bharatiya Nabhikiya Vidyut Nigam Limited) and developed with IGCAR (Indira Gandhi Centre for Atomic Research).

Stage-2 Action: Uses Plutonium-239 recovered from Stage-1 Pressurized Heavy Water Reactors (PHWRs). Fast neutrons convert Uranium-238 into more Plutonium-239 (producing more fuel than it consumes--"breeding").

Thorium Bridge: It will eventually blanket with Thorium-232, transmuting it to Uranium-233 to power Stage-3, tapping India's massive thorium reserves.

Policy Context:

The Union Budget 2025-26 launched the Nuclear Energy Mission targeting 100 GW of nuclear capacity by 2047.

SHANTI Act, 2025 (Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India Act, 2025) consolidates India's nuclear legal framework and allows limited private sector participation.

2. High-Profile Appointments & National Leadership (May 2026 Status)

NITI Aayog:

Chairman: Prime Minister Narendra Modi (Ex-officio).

Vice-Chairman: Ashok Kumar Lahiri (Appointed in April 2026, replacing Suman Bery; noted economist, former Chief Economic Advisor).

Interim CEO: Nidhi Chhibber (Assumed additional charge in February 2026 as B.V.R. Subrahmanyam's tenure ended).

Reserve Bank of India (RBI):

Governor: Sanjay Malhotra (26th Governor, in office since December 11, 2024).

SEBI (Securities and Exchange Board of India):

Chairperson: Madhabi Puri Buch.

Part 2: English Spell-Check & Vocabulary High-Yield Drill

NIELIT exams include multiple-choice questions where you must identify the correctly spelled word.

1. Highly Confused Spellings Matrix

Confused / Pitfall Spelling

Correct Spelling

Why it Traps

etiquette, etiquette

Etiquette

Single 't', double 't' in the middle, and ends in 'ette'.

paracial, palasial

Palatial

From "palace". Uses 't' instead of 'c'.

accomodate

Accommodate

Double 'c' and double 'm'.

homogeneous, homogenious

Homogeneous

Contains 'eou', not 'iou'.

paralleled, paralell

Parallel

Double 'l' first, then single 'l': P-a-r-a-l-l-e-l.

guarantee, guaranty

Guarantee

Ends with 'double e' for the verb, 'y' for the financial noun.

supercede

Supersede

The only word ending in -sede (from Latin supersedere), not -cede.

maintainance

Maintenance

Changes from "maintain" to main-te-nance.

occurance, occurrence

Occurrence

Double 'c', double 'r', and ends in -ence.

independent, independant
Independent
Ends with -ent, never -ant.

2. Vocabulary Tricks (Antonyms & Synonyms)

Elimination by Tone: Check if options have a positive (+), negative (-), or neutral (0) tone. If the question asks for a synonym of a positive word, eliminate negative options immediately.

Suffix Clues: Words ending in -less, -anti-, or -de- are usually negative, while -ful, -bene-, or -pro- are positive.

Root Association:

Crave $\rightarrow$ desire strongly (+)

Confront $\rightarrow$ face or match directly (neutral/assertive)

Accept $\rightarrow$ receive (+)

Deny $\rightarrow$ refuse (-)